

Machine Learning Meets Video Gaming

Supervisory Team

Ahmed M. A. Sayed, Queen Mary University of London, UK
Poonam Yadav, University of York, UK



UNIVERSITY
of York

Bio of Supervisory Team



Ahmed M. A. Sayed
QMUL, UK

- Designs & Develops Practical Systems
- Ph.D. (HKUST) in Comp Sci & Eng
- Blend of Industry and Research:
 - Senior Researcher (Huawei & DeepCloudAI)
 - Research Scientist (KAUST, KSA)

Current Research Themes:

- Federated Learning and ML systems
- Distributed and Networked Systems
- Cloud/Edge Computing and Data Analytics
- Computer Networks and SDN



Poonam Yadav
U of York, UK

- Trustworthy Privacy Preserved Systems
- Ph.D. (Imperial College) in Computing
- Blend of Industry and Research:
 - Researcher (IBM Research, USA)
 - Research Fellow (University of Cambridge, UK)

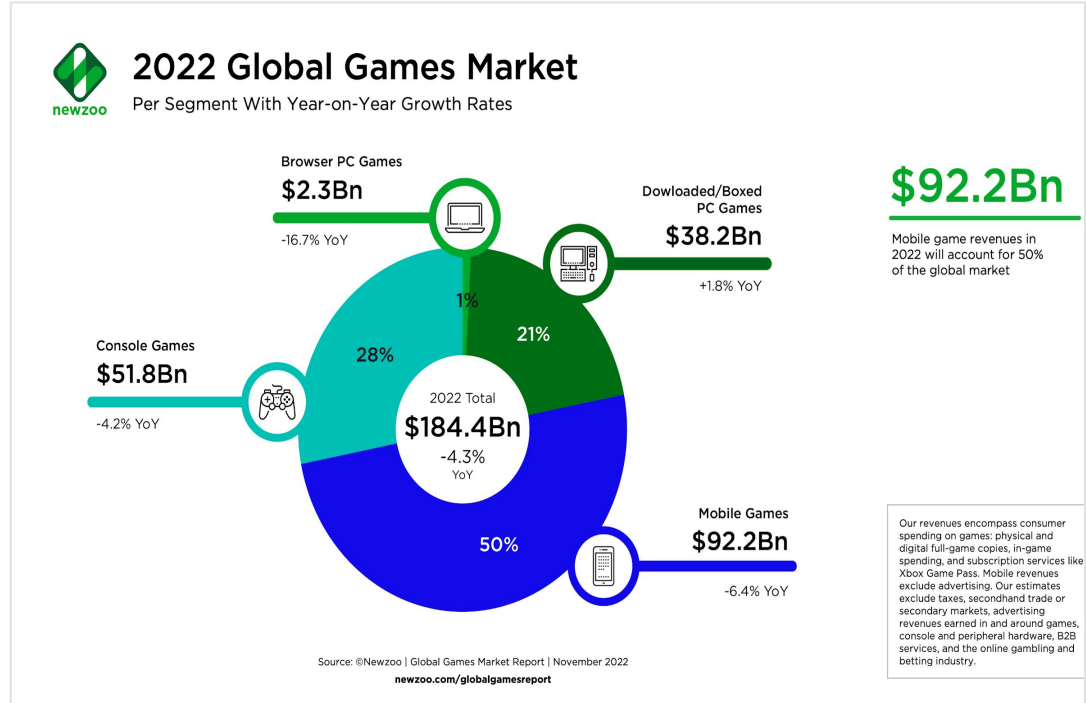
Current Research Themes:

- Federated Edge Computing
- Distributed computing for citizen science games
- Interoperability in Resilient Communication Networks
- Secure Data sharing in Edge/Cloud systems

Gaming Industry Slow Down by **up to -17%** - can we help?

Challenges

1. Fast testing and development for robust and resilient games with new immersive user experiences
2. User experience in internet connected games (Edge/Cloud computing)
3. Security and Privacy (Federated Learning)

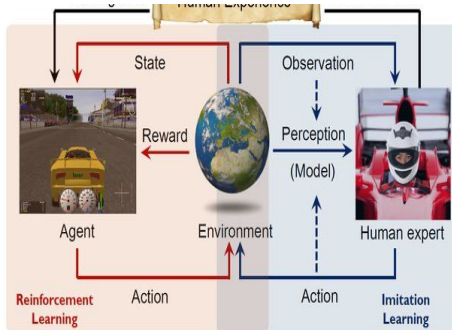


Topics of Interest (not conclusive)

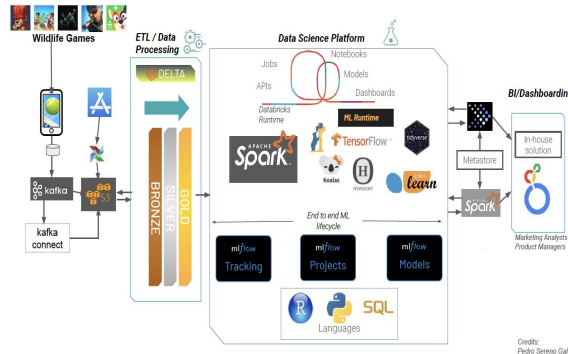
Main Themes:

- **AI/ML to automate testing & development of video games at scale**
- **Privacy-preserving collaborative learning for mobile game agents**

Reinforcement & imitation learning for testing in context of *distributed private game data*



User-centric automated and scalable ML development pipeline based on user-feedback loop



Leveraging Edge computing to enhance QoE via privacy-preserving federated learning for mobile games



<https://unity.com/products/machine-learning-agents>
<https://ai.googleblog.com/2021/03/leveraging-machine-learning-for-game.html>

Contact Us - See you on Discord

We are OPEN to other iGGi-Related Research Ideas

Reach out to us via email to initiate discussion/proposal

Ahmed M. A. Sayed

Lecturer (Asst. Prof.)

School of EECS, QMUL, UK

ahmed.sayed@qmul.ac.uk

<https://eecs.qmul.ac.uk/~ahmed>

Poonam Yadav

Lecturer (Asst. Prof.)

CS Dept., U of York, UK

poonam.yadav@york.ac.uk

<https://poonamyadav.net/>